Vertex Electronics

Technical Research Report

Predictive Maintenance Through Unsupervised Clustering

Empirical Analysis of the AI4I 2020 Predictive Maintenance Dataset

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Dataset:	UCI AI4I 2020 Predictive Maintenance Dataset
Method:	Standardized five-feature K-Means with post-hoc statistical analysis
Report version:	1.1
Date:	22 October 2026

This report documents an exploratory unsupervised-learning experiment. Failure labels were excluded from cluster fitting and used only in post-hoc analyses. The results are associations in a synthetic benchmark dataset, not validated forecasts of future failures.

Abstract

This study examines whether unsupervised clustering can identify interpretable operating-condition profiles in the AI4I 2020 Predictive Maintenance Dataset. Five continuous measurements—air temperature, process temperature, rotational speed, torque, and tool wear—were standardized and clustered using K-Means. Among the tested values $K = 2$ through $K = 6$, the three-cluster solution had the highest silhouette score (0.2622), the lowest Davies–Bouldin index (1.2437), and the highest Calinski–Harabasz index (3,831.00). Across ten seed-based runs, the mean pairwise adjusted Rand index (ARI) was 0.9975. In contrast, raw-feature and standardized-feature partitions had an ARI of 0.1855, showing substantial sensitivity to scaling. The observed failure rates were 5.24%, 2.17%, and 2.24% for Clusters 0, 1, and 2. The cluster–failure association was statistically significant, $\chi^2(2) = 67.81$, $p = 1.89 \times 10^{-15}$, but small in magnitude (Cramér's $V = 0.0823$). After adjustment for product type, the odds of failure in Cluster 0 were 2.48 times those in Cluster 1 (95% CI 1.93–3.21, $p < 0.001$). The overall cluster-by-type interaction test was not significant at the 5% level ($p = 0.0814$). These results are exploratory, in-sample associations in synthetic data; they do not establish causation or future predictive performance.

Keywords: predictive maintenance; unsupervised learning; K-Means; clustering; operating-condition profiles; machine-failure association.

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1 Introduction and research questions

Predictive maintenance uses equipment-condition information to inform inspection and maintenance decisions. Clustering can serve as an exploratory step when the goal is to describe operating regimes without using failure labels to define the regimes. It groups records by similarity in selected measurements; engineers can then examine whether groups differ in operating profiles or in retrospectively observed outcomes.

This report analyzes the UCI AI4I 2020 Predictive Maintenance Dataset, a synthetic benchmark intended to reflect selected characteristics of industrial predictive-maintenance data [1, 2]. The study is deliberately limited: it investigates operating-condition groupings and their association with observed failure labels. It does not train a supervised predictor, test temporal lead time, or evaluate prevented downtime.

The research questions are:

1. Can K-Means produce interpretable operating-condition groups from five standardized numerical measurements?
2. Which candidate cluster count provides the most defensible exploratory segmentation among $K = 2, \ldots, 6$ based on internal metrics and cluster size?
3. Are assignments reproducible across random initializations, and how sensitive are they to feature scaling?
4. Are cluster memberships associated with observed machine-failure labels after stratifying by and adjusting for product type?

2 Dataset and preprocessing

2.1 Dataset provenance and structure

The analysis used the local CSV file named `ai4i2020.csv`, corresponding to the dataset distributed by the UCI Machine Learning Repository (DOI: 10.24432/C5HS5C) [1]. UCI describes the data as synthetic and lists 10,000 observations and 14 columns. The cited dataset record identifies the license as Creative Commons Attribution 4.0 (CC BY 4.0). The dataset should therefore be attributed when redistributed; the actual downloaded file and its checksum should be retained in the project evidence package.

The supplied CSV contained 10,000 rows and 14 columns. The inspection found no missing values and no duplicate rows. The binary `Machine failure` label was positive for 339 records (3.39%). This low prevalence matters when interpreting failure-rate differences and would require special attention in any future supervised prediction task.

2.2 Clustering features and standardization

The primary feature matrix comprised five measurements: air temperature (K), process temperature (K), rotational speed (rpm), torque (Nm), and tool wear (min). Product type and every failure label were excluded from primary model fitting. Product type was brought back only for post-hoc stratification and adjustment; failure labels were examined only after cluster assignments were obtained.

Each continuous feature was standardized as

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \tag{1}$$

where x_{ij} denotes observation i on feature j, and μ_j and σ_j are the corresponding feature mean and standard deviation. Standardization places features expressed in different units on comparable scales

Table 1: Dataset fields grouped by their role in this experiment.

Role	Fields	Treatment in this study
Identifiers	`UDI`, `Product ID`	Excluded from clustering
Product attribute	`Type` (H, L, M)	Excluded from cluster fitting; used in stratified and adjusted analyses
Continuous measurements	Air temperature, process temperature, rotational speed, torque, tool wear	Five standardized features used to fit primary K-Means models
Outcome and failure-mode labels	`Machine failure`, `TWF`, `HDF`, `PWF`, `OSF`, `RNF`	Excluded from cluster fitting; only `Machine failure` used in the reported post-hoc outcome analysis

for Euclidean-distance-based clustering. The later sensitivity analysis shows that this preprocessing choice materially changes assignments.

3 Methodology

3.1 Analysis workflow

Figure 1 summarizes the analysis sequence. Outcome labels are absent from the model-fitting and internal-metric stages.

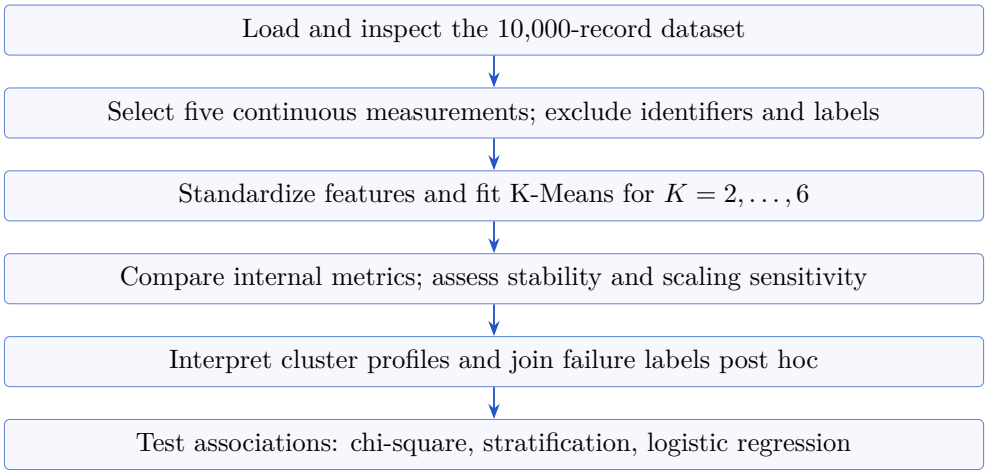


Figure 1: *Analysis workflow. Failure labels are excluded from model fitting and examined only after clusters are assigned.*

3.2 K-Means and configuration

K-Means partitions observations into K disjoint groups by minimizing the within-cluster sum of squared Euclidean distances [3]:

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2, \tag{2}$$

where C_k is cluster k and μ_k its centroid. The primary run used $K = 3$, `random_state=42`, and `n_init=10`. Cluster numbers are arbitrary labels; they do not intrinsically represent risk levels or a

temporal order.

3.3 Internal evaluation and K selection

Candidate K-Means models with $K = 2, 3, 4, 5, 6$ were compared using three internal metrics, calculated on the standardized feature matrix. These metrics are conventional measures of geometric cluster structure [4]:

- **Silhouette coefficient:** higher values indicate stronger average cohesion and separation; values near zero suggest overlap.
- **Davies–Bouldin index:** lower values indicate lower within-cluster scatter relative to centroid separation.
- **Calinski–Harabasz index:** higher values indicate greater between-cluster dispersion relative to within-cluster dispersion.

No one internal score proves that the selected partition reflects natural or operationally meaningful machine states. Selection therefore considers the joint metric pattern, cluster sizes, interpretability, and sensitivity.

3.4 Stability and scaling sensitivity

Stability was assessed by fitting K-Means with ten seeds (0 through 9), each using $K = 3$ and `n_init=10`. Agreement was summarized with the adjusted Rand index (ARI) for all 45 pairwise comparisons among the ten partitions. ARI values near 1 indicate close agreement in assignments, independently of the arbitrary cluster-label numbering [4].

For scaling sensitivity, a separate $K = 3$ K-Means model was fitted to the five unstandardized numerical features with the same primary random state and initialization count. Its partition was compared against the standardized partition using ARI. This is a sensitivity comparison, not a test that either representation is universally correct.

3.5 Post-hoc statistical analysis

Following clustering, `Machine failure` was compared across cluster assignments. The analyses comprised (1) cluster-specific observed failure rates, (2) a chi-square test of independence and Cramérś V, (3) failure-rate comparisons within each product type, (4) logistic regression adjusted for product type, and (5) a likelihood-ratio test comparing adjusted logistic models with and without cluster-by-type interactions.

In the adjusted model, Cluster 1 and Type L were the reference categories. The logistic coefficients were exponentiated to obtain odds ratios (ORs) with 95% Wald confidence intervals. The interaction test compared the model including `Cluster * Type` with the additive model. All these analyses are post hoc and in-sample. Statistical significance is not equivalent to a large effect, a causal effect, or demonstrated predictive utility.

4 Results

4.1 Dataset checks

The local CSV contained 10,000 observations, 14 columns, zero missing values, zero duplicate rows, and 339 positive machine-failure labels. The failure prevalence in the analysed file was 3.39%. No observations were removed during the documented checks.

4.2 Selecting the number of clusters

Table 2 reports the metrics obtained from the verified K-Means runs. For the tested range, $K = 3$ had the highest silhouette score, the lowest Davies–Bouldin index, and the highest Calinski–Harabasz index.

Table 2: Internal metrics for K-Means models fitted to the five standardized features.

K	Silhouette ↑	Davies–Bouldin ↓	Calinski–Harabasz ↑	Smallest cluster
2	0.2581	1.5163	3640.29	4,990
3	0.2622	1.2437	3831.00	2,053
4	0.2376	1.2859	3366.71	1,713
5	0.2250	1.3734	3194.03	1,062
6	0.2200	1.3332	3036.24	972

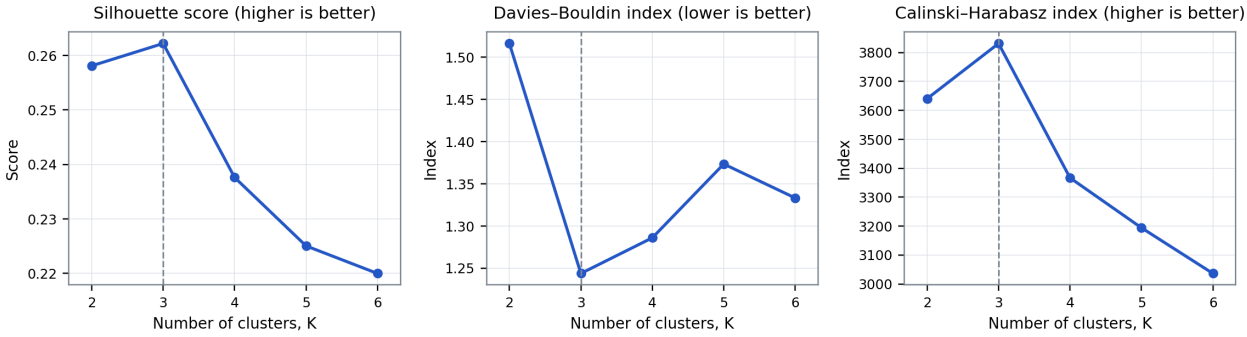


Figure 2: *Internal metrics across the tested K-Means cluster counts. The dashed line marks $K = 3$; higher is preferred for silhouette and Calinski–Harabasz, while lower is preferred for Davies–Bouldin.*

The selected model's silhouette score of 0.2622 indicates modest separation; the clusters overlap rather than forming three sharply separated groups. K=3 is retained as a practical exploratory segmentation within the tested range, not as proof that the data contain exactly three natural operating regimes.

4.3 Cluster sizes and operating profiles

The selected solution assigned 3,930 records to Cluster 0, 4,017 to Cluster 1, and 2,053 to Cluster 2. Table 3 reports means in original measurement units so that profiles can be interpreted in engineering terms.

Table 3: Mean operating measurements by cluster, in original units.

Feature	Cluster 0	Cluster 1	Cluster 2
Air temperature (K)	301.72	298.26	300.12
Process temperature (K)	311.25	308.74	310.10
Rotational speed (rpm)	1469.81	1474.14	1797.26
Torque (Nm)	43.64	43.26	26.57
Tool wear (min)	110.36	104.62	109.85

Cluster 0 has the highest mean air and process temperatures. Cluster 1 has the lowest mean temperatures and the lowest observed failure rate. Cluster 2 has substantially higher mean rotational speed and lower mean torque than the other groups. Figure 3 expresses each cluster's feature means in dataset-standard-deviation units so that features on different measurement scales can be compared in

one graphic. These profiles describe operating conditions; they should not be interpreted as a universal risk ranking.

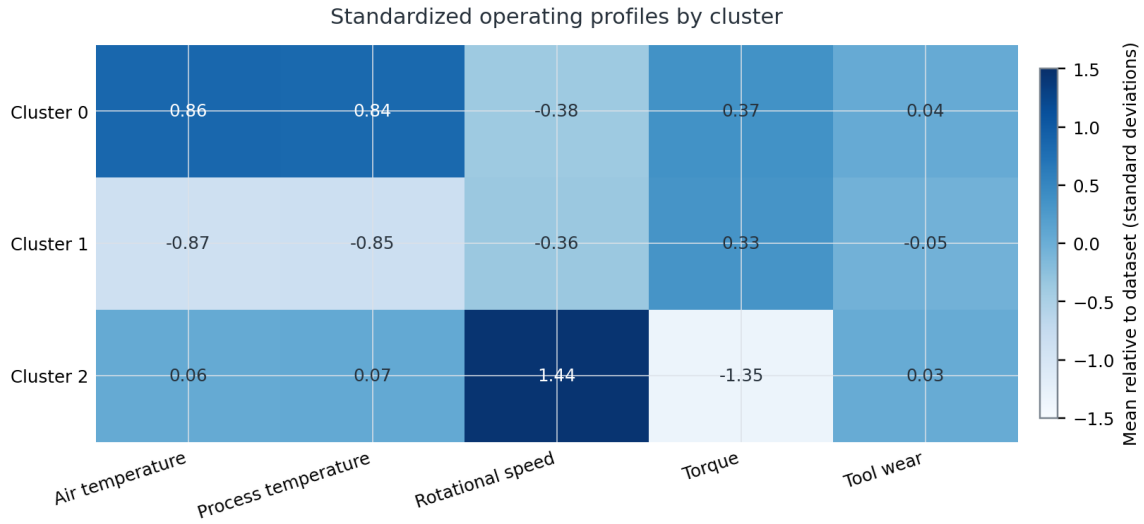


Figure 3: *Cluster-level mean operating profiles in standardized units. Values are means relative to each feature's full-dataset mean and standard deviation; the accompanying table gives original units.*

4.4 Stability and feature-scaling sensitivity

Across ten runs, the 45 pairwise ARI comparisons had a mean of 0.9975, a minimum of 0.9920, and a maximum of 1.0000. This indicates that assignments were highly reproducible under the same features, standardization method, cluster count, and initialization settings. It does not establish that the selected partition is uniquely correct or predictive.

The partition obtained without standardization had an ARI of 0.1855 relative to the standardized partition. The raw-feature solution had cluster sizes of 5,664, 3,696, and 640. Because cluster labels are arbitrary, these sizes are reported as a size profile rather than as a one-to-one matching with the standardized clusters. The low agreement indicates that cluster assignments are highly sensitive to scaling. Standardization is defensible because the features use different units and numeric ranges, but the conclusion is conditional on that choice.

4.5 Observed failure rates by cluster

Table 4 gives the post-hoc failure counts and rates for the selected standardized model.

Table 4: Observed machine-failure counts and rates by cluster.

Cluster	Records	Failures	No failure	Failure rate
0	3,930	206	3,724	5.24%
1	4,017	87	3,930	2.17%
2	2,053	46	2,007	2.24%

The chi-square test of independence yielded $\chi^2(2) = 67.8088$ and $p = 1.8858 \times 10^{-15}$. Cramér's V was 0.0823, indicating a small association despite the small p-value. The result supports a statistical association between cluster membership and observed machine-failure labels in this dataset; it does not establish causality or operational value.

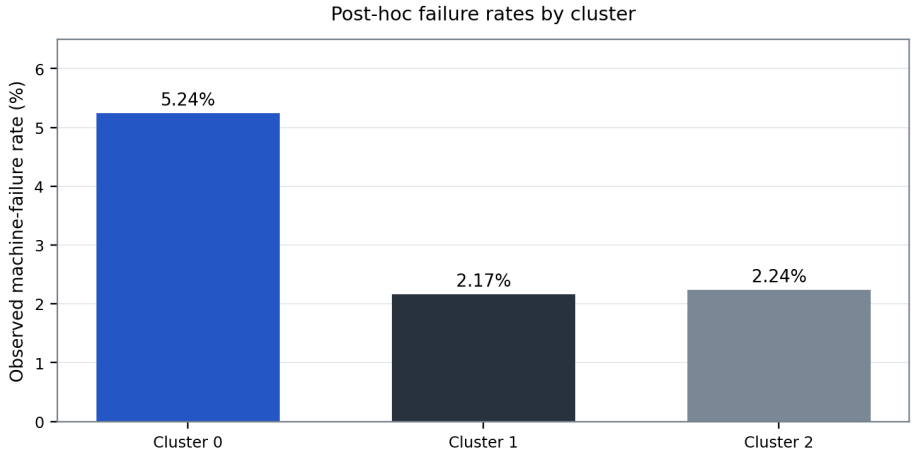


Figure 4: *Observed machine-failure label rate by cluster. These are in-sample descriptive rates, not out-of-sample prediction metrics.*

4.6 Stratification by product type

To assess whether product-type composition alone could explain the cluster-level failure-rate pattern, failure rates were compared within each type. Table 5 shows that Cluster 0 has the highest observed failure rate within H, L, and M products. The percentages of product types within clusters were also similar (Table 6).

Table 5: Observed failure rates within product type and cluster.

Product type	Cluster 0	Cluster 1	Cluster 2
H	4.43%	0.91%	0.50%
L	5.99%	2.73%	2.24%
M	4.00%	1.51%	2.82%

Table 6: Product-type composition within each standardized cluster (percent).

Cluster	Type H	Type L	Type M
0	9.19%	60.28%	30.53%
1	11.00%	59.27%	29.72%
2	9.74%	60.89%	29.37%

The descriptive pattern within product types makes it less likely that the overall association is explained solely by type composition. However, stratification does not eliminate all potential confounding and is not a causal test.

4.7 Logistic regression adjusted for product type

The adjusted logistic model used Cluster 1 and Type L as reference categories. Table 7 presents the cluster and type comparisons.

After adjustment for product type, the estimated odds of failure in Cluster 0 were 2.48 times those in Cluster 1 (95% CI 1.93–3.21, $p < 0.001$). Cluster 2 did not differ significantly from Cluster 1 (OR 1.03, 95% CI 0.72–1.47, $p = 0.8847$). These ORs compare odds, not probabilities or risk ratios, and describe the analysed records rather than future events.

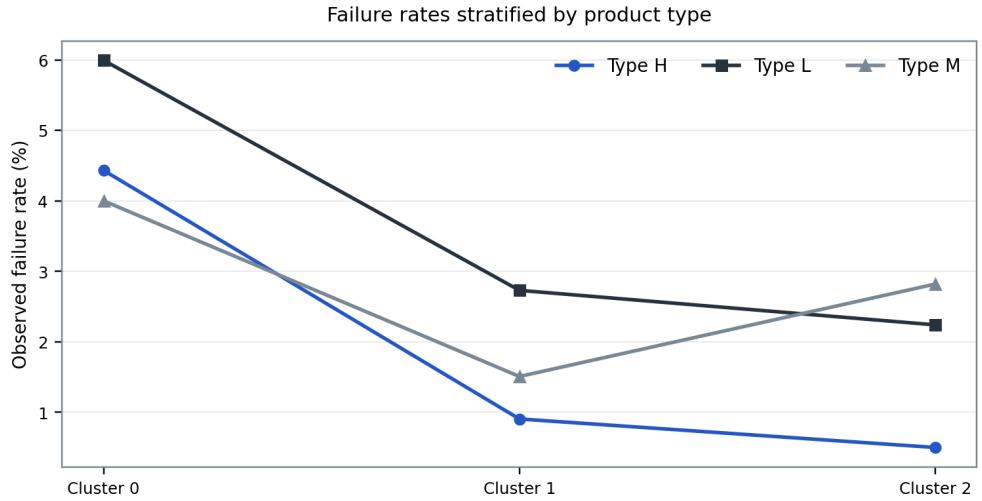


Figure 5: *Observed failure rates stratified by product type and cluster. The plot is descriptive and does not establish causation.*

Table 7: Product-type-adjusted logistic regression estimates. Reference categories: Cluster 1 and Type L.

Comparison	Adjusted OR	95% CI	p-value
Cluster 0 vs. Cluster 1	2.4848	1.9259–3.2058	< 0.001
Cluster 2 vs. Cluster 1	1.0271	0.7155–1.4744	0.8847
Type H vs. Type L	0.5404	0.3437–0.8497	0.0077
Type M vs. Type L	0.6936	0.5375–0.8950	0.0049

4.8 Cluster-by-type interaction test

A likelihood-ratio test compared the adjusted additive logistic model against a model including cluster-by-product-type interaction terms. The statistic was 8.2934 with 4 degrees of freedom ($p = 0.0814$). At the 5% threshold, the overall interaction was not statistically significant. The analysis therefore does not provide sufficient evidence that the cluster–failure association differs across product types; it does not prove the associations are identical.

One individual interaction coefficient (Cluster 2 × Type M) had a nominal p-value of 0.0407. However, the overall interaction test was not significant, and several interaction terms were examined. The individual result should therefore be treated as exploratory and does not override the overall test. Full coefficient estimates are reported in Appendix A.

5 Discussion and engineering implications

The experiment supports three restrained conclusions. First, K-Means produced three reproducible operating-condition groups under the selected standardized feature representation. Second, the groups differed in their operating profiles, especially temperature for Cluster 0 and rotational speed/torque for Cluster 2. Third, cluster membership was associated with observed failure labels, and the association for Cluster 0 remained statistically significant after adjusting for product type.

The finding may be useful as a hypothesis-generation step. For example, engineers could inspect whether records with comparable temperature profiles are associated with particular failure mechanisms in real, time-stamped telemetry. Yet the present dataset and design do not show that temperature causes failure, that these clusters would identify failure sufficiently early, or that a cluster-based

intervention would reduce downtime. Those claims require a separate, prospective or time-aware evaluation.

Two findings should be read together. The mean pairwise ARI of 0.9975 shows high repeatability under a fixed feature and scaling design. The raw-versus-standardized ARI of 0.1855 shows that the design itself substantially affects the partition. A model can be stable under repeated runs while remaining sensitive to a preprocessing decision. Both results are important to report.

Finally, the small Cramér's V (0.0823) qualifies the interpretation of the very small chi-square p-value. With 10,000 records, a small association may be statistically detectable, but effect size and potential decision value are different questions. The study did not test operational thresholds, alerts, false alarms, missed failures, or financial outcomes.

6 Limitations and threats to validity

1. **Synthetic benchmark.** The dataset reflects an intentionally synthetic approximation of industrial data and may not represent real-world noise, equipment variation, maintenance policies, or event prevalence [1, 2].
2. **Modest cluster separation.** The selected silhouette score of 0.2622 indicates overlap. The clusters should be interpreted as exploratory groupings, not sharply separated machine states.
3. **Scaling sensitivity.** The raw-versus-standardized ARI of 0.1855 shows that membership depends substantially on feature scaling. Another defensible transformation or weighting strategy may create a different partition.
4. **No temporal validation.** The analysis does not test whether cluster assignments precede failure events, remain stable over time, or generalize to unseen machines.
5. **No predictive evaluation.** No time-aware held-out test, supervised classifier, calibration analysis, precision-recall analysis, or lead-time evaluation was performed.
6. **Small association magnitude.** Cramér's $V = 0.0823$ indicates a small cluster–failure association. A low p-value alone is not evidence of operational usefulness.
7. **Residual confounding.** Adjustment for product type does not control all potentially relevant variables and does not establish causation.
8. **Restricted model comparison.** The verified experimental evidence reported here covers K-Means for $K = 2$ through $K = 6$. It does not claim a verified Ward-linkage or Gaussian-mixture comparison.
9. **Reproduction dependencies.** Exact regeneration requires the same source CSV, compatible package versions, the script and parameters included in the project package. The source package documents how to regenerate the analyses; it does not embed the dataset itself.

7 Recommendations and future work

The next stage should prioritize validation rather than adding more in-sample tests without a decision-oriented reason.

1. **Validate using real telemetry.** Gather time-stamped sensor measurements, machine identifiers, maintenance logs, and failure-event times from equipment relevant to the intended use.
2. **Validate across machines and time.** Assess profile stability across different machines and future periods. If prediction is pursued, avoid random splits that allow temporal leakage.
3. **Review with maintenance specialists.** Determine whether the operating profiles correspond to meaningful regimes and whether they imply feasible inspections or diagnostic checks.

4. **Measure operational value.** Track alert workload, inspection yield, missed failures, downtime and maintenance cost before recommending deployment.
5. **Use a separate supervised evaluation for forecasting.** If the goal becomes failure prediction, evaluate a supervised model using time-separated held-out data and class-imbalance-aware metrics, such as precision, recall, precision–recall AUC, calibration and lead time.
6. **Document robustness comparisons.** Future model comparisons should use documented code, a consistent feature-processing policy, relevant internal and external validation, and reported random seeds. Do not include methods in the report unless their results are evidenced.

8 Conclusion

K-Means with three clusters provided a highly reproducible exploratory segmentation of five standardized operating measurements in the AI4I 2020 dataset. Among the tested values $K = 2$ through $K = 6$, the selected solution had a silhouette score of 0.2622, Davies–Bouldin index of 1.2437, and Calinski–Harabasz index of 3,831.00. Cluster 0 had the highest observed failure rate (5.24%), and its adjusted odds of failure were higher than Cluster 1 after controlling for product type (OR 2.48, 95% CI 1.93–3.21). The overall cluster-by-type interaction did not reach statistical significance at the 5% threshold.

The findings support an association between operating-condition groupings and observed failure labels in this synthetic dataset. They do not establish a predictive-maintenance capability, causation, or a reduction in downtime. Modest cluster separation, strong scaling sensitivity, the synthetic data source, and the absence of temporal validation limit the conclusions. The next defensible step is validation on real, time-stamped equipment telemetry.

A Full interaction-model estimates

Table 8 provides the interaction-model odds ratios and confidence intervals. Terms for Cluster 0 and Cluster 2 are relative to Cluster 1 when Type L is the reference type. Interaction terms represent departures from the additive model on the log-odds scale; they should not be interpreted as standalone cluster effects for every type.

Table 8: Full logistic model with cluster-by-type interactions. Reference categories: Cluster 1 and Type L.

Term	Coefficient	OR	95% CI for OR	p-value
Intercept	-3.5732	0.0281	0.0219–0.0359	< 0.001
Cluster 0 vs. 1	0.8206	2.2719	1.6844–3.0644	< 0.001
Cluster 2 vs. 1	-0.2028	0.8164	0.5214–1.2784	0.3753
Type H vs. L	-1.1227	0.3254	0.1179–0.8977	0.0301
Type M vs. L	-0.6063	0.5454	0.3221–0.9235	0.0241
Cluster 0 × Type H	0.8043	2.2352	0.7117–7.0206	0.1684
Cluster 2 × Type H	-0.3946	0.6740	0.0715–6.3499	0.7303
Cluster 0 × Type M	0.1808	1.1982	0.6419–2.2367	0.5702
Cluster 2 × Type M	0.8422	2.3215	1.0365–5.1999	0.0407

The model converged on all 10,000 observations. The likelihood-ratio comparison of additive and interaction models was $\chi^2(4) = 8.2934$, $p = 0.0814$. The overall test is the primary inferential result for whether interaction terms jointly improve model fit. The nominal p-value for one individual interaction coefficient should be interpreted cautiously in the presence of multiple terms.

B Reproducibility statement and data notes

The primary analysis used the five continuous features listed in Section 2, standardized using `StandardScaler`. The selected K-Means fit used $K = 3$, `random_state=42`, and `n_init=10`; the model-count comparison covered $K = 2, 3, 4, 5, 6$. Stability analysis used seeds 0–9, producing 45 pairwise ARI comparisons. The chi-square and regression analyses were fitted after clustering; no failure label was used in fitting the K-Means models.

The companion `analysis/analyze.py` script regenerates result tables and figures from a local copy of `ai4i2020.csv`. The CSV is not bundled in the source package; the data README provides the official UCI source, citation and access instructions. The installed software versions used for the validation run are recorded in `results/analysis_summary.json`. For a public release, retain the exact data version and checksum alongside the analysis output.

References

[1] UCI Machine Learning Repository. *AI4I 2020 Predictive Maintenance Dataset*. 2020. DOI: 10.24432/C5HS5C. Dataset record and license information accessed 22 October 2026.

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[4] Scikit-learn developers, "Clustering performance evaluation," *scikit-learn documentation*. Available: https://scikit-learn.org/stable/modules/clustering.html. Accessed 22 October 2026.